

ROBOTICS (12/02/2020)

Matteo Matteucci, Simone Mentasti

The exam will be graded IFF the following recommendations have been taken into account:

- Write clearly so that the teacher can easily understand your answers
- Write your name, surname, and student id on each sheet you deliver for evaluation
- For each exercise/question report clearly the number and sub-number (if present)
- You are not allowed to use any programmable device (e.g., smartphone, calculator, etc.)
- You can use pen or pencil, paper will be provided, you cannot use notes or books

Exercise 1 – Sensors and Actuators [2+2+2+2 points]

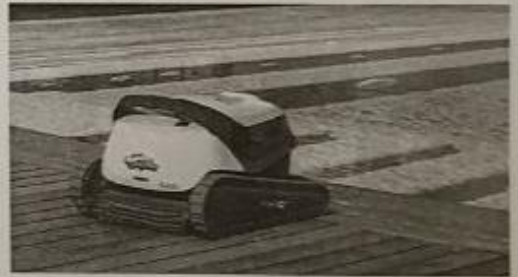
Considering the classical sensors and actuators used in mobile robots, answer the following:

- (a) What is the difference between active and passive sensors? Make one example for each category.
- (b) What is the difference between proprioceptive and exteroceptive sensors? Make one example for each category.
- (c) What is the difference between brushed and brushless motors? Which one would you prefer for a domestic vacuum cleaner?
- (d) How do you measure the number of revolutions in a brushed motor? Provide a detailed description.

Exercise 2 – Robot Odometry [2+2+2 point]

The Dolphin S200 is an electric robot for the cleaning of swimming pools. As you can see from the picture it has a peculiar kinematics. With reference to its kinematics answer the following questions:

- a) What are direct and inverse kinematics and what are their typical uses?
- b) Provide the approximated direct and inverse kinematics for the Dolphin S200 swimming pool cleaner.
- c) Odometry is the integration of direct kinematics to compute the path traversed by the robot. Provide the basic (approximated) odometric formulas for a robot with the kinematics of the Dolphin S200. In what sense they are approximated? How can be improved?



Exercise 3 – Robot Local Path Planning [1+1+1+2]

Let's consider the problem of obstacle avoidance, a.k.a., local path planning, and the algorithms to realize it. Provide an answer to the following questions:

- a) What is local path planning goal? What should be its frequency with respect to global path planning?
- b) Do you need global path planning for the swimming pool cleaning task in Exercise 2? Why?
- c) Which sensors do you need for the swimming pool cleaning task in Exercise 2? Why?
- d) Which local path planning algorithm would you chose for the swimming pool cleaner of Exercise 2? Why?

Exercise 4 – Robot Operating System [3 points]

Answer the following questions:

- a) What is ROS?
- b) What are nodes?
- c) What are topics?

Exercise 5 – Localization and Mapping [1+2+2]

Let's consider the Simultaneous Localization and Mapping problem; answer the following questions:

- a) Provide the derivation of the recursive Bayes Filter
- b) How does the Monte Carlo Localization algorithm implement the recursive Bayes Filter for Localization?
- c) How does the EKF SLAM algorithm implement the recursive Bayes Filter for online Simultaneous Localization and Mapping?